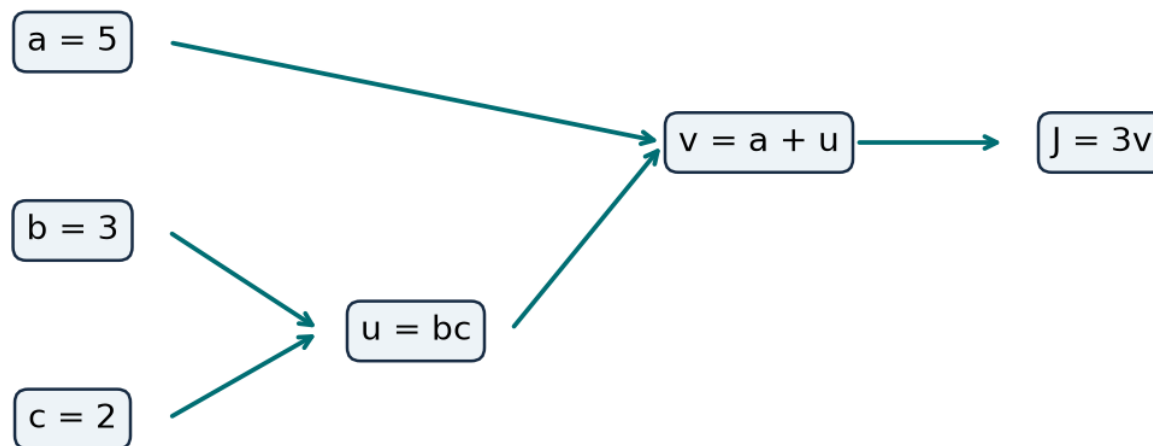


WORKED EXAMPLE

Break one expression into small steps

Lecture 2, page 27: a computation graph records which values depend on which.

$$J(a, b, c) = 3(a + bc), \quad a = 5, \quad b = 3, \quad c = 2$$



Arrows show the forward flow of values. u and v are intermediate results. We compute values forward and sensitivities backward.

WORKED EXAMPLE

Forward pass: save every node value

Compute a node only after its inputs are known.

$$u = bc = 3 \times 2 = 6$$

$$v = a + u = 5 + 6 = 11$$

$$j = 3v = 3 \times 11 = 33$$

Saving u and v avoids repeatedly expanding the full expression. Saving b and c matters because the backward calculation needs their values.

A node value answers "what number did we compute?" A derivative answers "how sensitive is the final result to a small change?"

WORKED EXAMPLE

Local derivatives: inspect one operation

Hold the other inputs fixed when differentiating an operation.

$$J = 3v \Rightarrow \frac{\partial J}{\partial v} = 3$$

$$v = a + u \Rightarrow \frac{\partial v}{\partial a} = 1, \quad \frac{\partial v}{\partial u} = 1$$

$$u = bc \Rightarrow \frac{\partial u}{\partial b} = c = 2, \quad \frac{\partial u}{\partial c} = b = 3$$

An addition node passes sensitivity to each input unchanged: multiply by 1.

A multiplication node uses the OTHER input as its local derivative. Changing b affects u in proportion to c .

WORKED EXAMPLE

Backward pass: multiply along each path

The chain rule multiplies upstream sensitivity by the next local derivative.

$$\frac{\partial J}{\partial u} = \frac{\partial J}{\partial v} \frac{\partial v}{\partial u} = 3(1) = 3$$

$$\frac{\partial J}{\partial a} = \frac{\partial J}{\partial v} \frac{\partial v}{\partial a} = 3(1) = 3$$

$$\frac{\partial J}{\partial b} = \frac{\partial J}{\partial u} \frac{\partial u}{\partial b} = 3(2) = 6$$

$$\frac{\partial J}{\partial c} = \frac{\partial J}{\partial u} \frac{\partial u}{\partial c} = 3(3) = 9$$

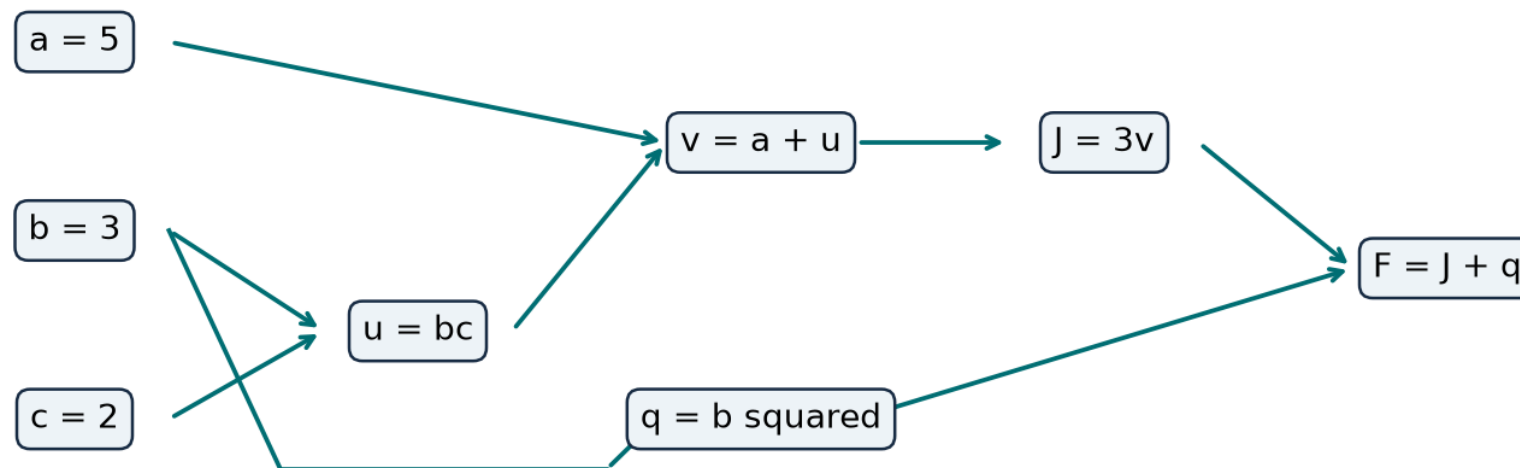
Near these inputs, increasing b by 0.01 increases J by about 6 times 0.01 = 0.06. The derivative and the node value u both happen to equal 6 here; they mean different things.

WORKED EXAMPLE

A shared input can affect two paths

Original study extension: add b squared to the lecture expression.

$$q = b^2 = 3^2 = 9, \quad F = J + q = 33 + 9 = 42$$



b now influences F through u and through q . Both effects matter. F and q are new extension nodes, not nodes from the lecture slide.

WORKED EXAMPLE

At a branch, add the path contributions

Changing b changes both J and q , so their effects on F add.

$$\frac{\partial F}{\partial b} = \frac{\partial F}{\partial J} \frac{\partial J}{\partial b} + \frac{\partial F}{\partial q} \frac{\partial q}{\partial b}$$

$$= (1)(6) + (1)(2b) = 6 + 2(3) = 12$$

$$\frac{\partial F}{\partial a} = 1(3) = 3, \quad \frac{\partial F}{\partial c} = 1(9) = 9$$

Multiply derivatives **ALONG** a path; add contributions **FROM** different paths. Do not discard one branch or multiply the two branch contributions.

We have computed sensitivities only. A parameter update would be a separate step using a learning rate.

INDEPENDENT PRACTICE

Your turn: include a negative input

Use the lecture graph first, then add the branching extension.

$$a = 1, \quad b = 2, \quad c = -1$$

1. Calculate $u = bc$, $v = a + u$, and $J = 3v$.
 2. Calculate the three derivatives of J with respect to a , b and c .
 3. Add $q = b$ squared and $F = J + q$. Calculate F and its derivative with respect to b .
- Show each local derivative and each path contribution, not just the final answers.

WORKED ANSWER

Practice answer: follow values and paths

The negative c makes the J path contribution for b negative.

$$u = 2(-1) = -2, \quad v = 1 + (-2) = -1, \quad J = 3(-1) = -3$$

$$\frac{\partial J}{\partial a} = 3(1) = 3, \quad \frac{\partial J}{\partial b} = 3(1)(-1) = -3$$

$$\frac{\partial J}{\partial c} = 3(1)(2) = 6$$

$$q = 2^2 = 4, \quad F = -3 + 4 = 1$$

$$\frac{\partial F}{\partial b} = 1(-3) + 1(2b) = -3 + 2(2) = 1$$

The branches partly cancel. Common mistakes: using b instead of c for the derivative of bc with respect to b, or confusing $F = 1$ with its derivative also being 1 here.